THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE UNIT TEST SET C

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Cell division in which one parent cell gives two identical daughter cells is: 1

(a) fertilisation **(b)** meiosis **(c)** osmosis **(d)** mitosis

2. The energy currency of the cell is: 1

(a) ATP **(b)** glucose only **(c)** RNA **(d)** DNA

3. Grass regrows after being cut because of: 1

(a) intercalary meristem **(b)** lateral meristem
(c) apical meristem **(d)** collenchyma

4. The number of pairs of ribs in the human rib cage is: 1

(a) 12 **(b)** 24 **(c)** 14 **(d)** 10

5. The two-part naming system was introduced by: 1

(a) Darwin **(b)** Whittaker **(c)** Carolus Linnaeus **(d)** Aristotle

6. Lichens are a symbiotic association between: 1

(a) two fungi **(b)** a bacterium and an alga
(c) a moss and a fern **(d)** an alga and a fungus

7. In a scientific name, the genus name is written: 1

(a) in the local language **(b)** in small letters only
(c) after the species **(d)** with a capital first letter, in italics

8. Menstruation occurs when: 1

(a) fertilisation takes place **(b)** the sperm dies
(c) pregnancy begins **(d)** the egg is not fertilised

9. The duration of the menstrual cycle is usually about: 1

(a) 90 days **(b)** 7 days **(c)** 48 days **(d)** 28 days

10. Potato reproduces vegetatively by: 1

(a) tubers (eyes) **(b)** leaves **(c)** seeds **(d)** spores

11. Winds are caused by: 1

(a) tides only **(b)** uneven heating of the Earth's surface
(c) rotation of the Moon **(d)** volcanic eruptions

12. A wavelength of 400 nm in metres is: 1

(a) 400×10^{-9} m **(b)** 400×10^{-3} m **(c)** 400×10^{-12} m **(d)** 400×10^{-6} m

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. State the cell theory in brief. 2

14. What is transpiration? State its importance. 2

15. What is classification? Why is it needed? 2

16. What are spores? Why are they produced in large numbers? 2

17. Why do polar regions receive less heat than the equator? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Why do plant cells need a cell wall while animal cells do not? 3

19. Why can the stem of a young mango sapling bend while a coconut husk is tough? 3

20. State the advantages of seeds and fruits in angiosperms. 3

21. State two roles of the atmosphere in supporting life. 3

22. What is the role of the Golgi apparatus? 3

OR

Describe how the cell membrane is a fluid mosaic. 3

23. What would happen if a plant could not transport food from leaves to roots? 3

OR

Describe the structure and functions of the epidermis in a leaf and root. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) What is the epidermis? 5
(ii) What are stomata and guard cells?
(iii) Explain transpiration and its significance.

OR

(i) Describe the structure of parenchyma, collenchyma and sclerenchyma. 5
(ii) Give one function of each.

25. **Case study:** Wind pollination has low success rate for seed formation compared to insect pollination. 5
(i) Why do wind-pollinated plants produce more pollen?
(ii) Describe the pollen of an insect-pollinated flower.
(iii) Why do flowers produce nectar?
(iv) Name a bird-pollinated plant.

OR

(i) Describe the cutting technique. 5
(ii) What is grafting?
(iii) Mention three advantages of vegetative propagation.

END OF QUESTION PAPER — Total: 50 Marks

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